In [2]:
```
!pip install numpy
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: numpy in C:\Users\VASANT PATIL homeuse\AppData\Roa
ming\Python\Python313\site-packages (2.5.3)
```

In [3]:
```
!pip install pandas
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pandas in C:\Users\VASANT PATIL homeuse\AppData\Ro
aming\Python\Python313\site-packages (3.0.6)
Requirement already satisfied: numpy>=1.26.0 in C:\Users\VASANT PATIL homeuse\App
Data\Roaming\Python\Python313\site-packages (from pandas) (2.5.3)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\Users\VASANT PATIL ho
meuse\AppData\Roaming\Python\Python313\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: tzdata in C:\Users\VASANT PATIL homeuse\AppData\Ro
aming\Python\Python313\site-packages (from pandas) (2026.4)
Requirement already satisfied: six>=1.5 in C:\Users\VASANT PATIL homeuse\AppData
\Roaming\Python\Python313\site-packages (from python-dateutil>=2.8.2->pandas) (1.
17.0)
```

In [4]:
```
!pip install matplotlib
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: matplotlib in C:\Users\VASANT PATIL homeuse\AppDat
a\Roaming\Python\Python313\site-packages (3.11.2)
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\VASANT PATIL homeuse
\AppData\Roaming\Python\Python313\site-packages (from matplotlib) (1.4.0)
Requirement already satisfied: cycler>=0.10 in C:\Users\VASANT PATIL homeuse\AppD
ata\Roaming\Python\Python313\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in C:\Users\VASANT PATIL homeuse
\AppData\Roaming\Python\Python313\site-packages (from matplotlib) (4.66.1)
Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\VASANT PATIL homeuse
\AppData\Roaming\Python\Python313\site-packages (from matplotlib) (1.5.1)
Requirement already satisfied: numpy>=1.25 in C:\Users\VASANT PATIL homeuse\AppDa
ta\Roaming\Python\Python313\site-packages (from matplotlib) (2.5.3)
Requirement already satisfied: packaging>=20.0 in C:\Users\VASANT PATIL homeuse\A
ppData\Roaming\Python\Python313\site-packages (from matplotlib) (26.3)
Requirement already satisfied: pillow>=9 in C:\Users\VASANT PATIL homeuse\AppData
\Roaming\Python\Python313\site-packages (from matplotlib) (12.3.0)
Requirement already satisfied: pyparsing>=3 in C:\Users\VASANT PATIL homeuse\AppD
ata\Roaming\Python\Python313\site-packages (from matplotlib) (3.3.3)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\VASANT PATIL home
use\AppData\Roaming\Python\Python313\site-packages (from matplotlib) (2.9.0.post
0)
Requirement already satisfied: six>=1.5 in C:\Users\VASANT PATIL homeuse\AppData
\Roaming\Python\Python313\site-packages (from python-dateutil>=2.7->matplotlib)
(1.17.0)
```

In [5]:
```
!pip install seaborn
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: seaborn in C:\Users\VASANT PATIL homeuse\AppData\R
oaming\Python\Python313\site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in C:\Users\VASANT PATIL home
use\AppData\Roaming\Python\Python313\site-packages (from seaborn) (2.5.3)
Requirement already satisfied: pandas>=1.2 in C:\Users\VASANT PATIL homeuse\AppDa
ta\Roaming\Python\Python313\site-packages (from seaborn) (3.0.6)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in C:\Users\VASANT PATIL h
omeuse\AppData\Roaming\Python\Python313\site-packages (from seaborn) (3.11.2)
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\VASANT PATIL homeuse
\AppData\Roaming\Python\Python313\site-packages (from matplotlib!=3.6.1,>=3.4->se
aborn) (1.4.0)
Requirement already satisfied: cycler>=0.10 in C:\Users\VASANT PATIL homeuse\AppD
ata\Roaming\Python\Python313\site-packages (from matplotlib!=3.6.1,>=3.4->seabor
n) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in C:\Users\VASANT PATIL homeuse
\AppData\Roaming\Python\Python313\site-packages (from matplotlib!=3.6.1,>=3.4->se
aborn) (4.66.1)
Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\VASANT PATIL homeuse
\AppData\Roaming\Python\Python313\site-packages (from matplotlib!=3.6.1,>=3.4->se
aborn) (1.5.1)
Requirement already satisfied: packaging>=20.0 in C:\Users\VASANT PATIL homeuse\A
ppData\Roaming\Python\Python313\site-packages (from matplotlib!=3.6.1,>=3.4->seab
orn) (26.3)
Requirement already satisfied: pillow>=9 in C:\Users\VASANT PATIL homeuse\AppData
\Roaming\Python\Python313\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn)
(12.3.0)
Requirement already satisfied: pyparsing>=3 in C:\Users\VASANT PATIL homeuse\AppD
ata\Roaming\Python\Python313\site-packages (from matplotlib!=3.6.1,>=3.4->seabor
n) (3.3.3)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\VASANT PATIL home
use\AppData\Roaming\Python\Python313\site-packages (from matplotlib!=3.6.1,>=3.4-
>seaborn) (2.9.0.post0)
Requirement already satisfied: tzdata in C:\Users\VASANT PATIL homeuse\AppData\Ro
aming\Python\Python313\site-packages (from pandas>=1.2->seaborn) (2026.4)
Requirement already satisfied: six>=1.5 in C:\Users\VASANT PATIL homeuse\AppData
\Roaming\Python\Python313\site-packages (from python-dateutil>=2.7->matplotlib!=
3.6.1,>=3.4->seaborn) (1.17.0)
```

In [6]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt # visualizing data
%matplotlib inline
import seaborn as sns
```

In [8]:
```
# import csv file
df = pd.read_csv('Diwali Sales Data.csv', encoding= 'unicode_escape')
```

In [9]:
```
df.shape
```

Out[9]:
```
(11251, 15)
```

In [10]:
```
df.head()
```

Out[10]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>User_ID</th> <th>Cust_name</th> <th>Product_ID</th> <th>Gender</th> <th>Age Group</th> <th>Age</th> <th>Marital_Status</th> <th>State</th> <th>Zone</th> <th>Occupation</th> <th>Product_Category</th> <th>Orders</th> <th>Amount</th> <th>Status</th> <th>unnamed1</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare	Au					
1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt	Au					
1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile	Au					
1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction	Au					
1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing	Au					

In [11]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   User_ID           11251 non-null  int64  
 1   Cust_name         11251 non-null  str    
 2   Product_ID        11251 non-null  str    
 3   Gender            11251 non-null  str    
 4   Age Group         11251 non-null  str    
 5   Age               11251 non-null  int64  
 6   Marital_Status    11251 non-null  int64  
 7   State             11251 non-null  str    
 8   Zone              11251 non-null  str    
 9   Occupation        11251 non-null  str    
 10  Product_Category  11251 non-null  str    
 11  Orders            11251 non-null  int64  
 12  Amount            11239 non-null  float64
 13  Status            0 non-null      float64
 14  unnamed1          0 non-null      float64
dtypes: float64(3), int64(4), str(8)
memory usage: 1.3 MB
```

In [12]:
```
#drop unrelated/blank columns
df.drop(['Status', 'unnamed1'], axis=1, inplace=True)
```

In [13]:
```
#check for null values
pd.isnull(df).sum()
```

Out[13]:
```
User_ID             0
Cust_name           0
Product_ID          0
Gender              0
Age Group           0
Age                 0
Marital_Status      0
State               0
Zone                0
Occupation          0
Product_Category    0
Orders              0
Amount             12
dtype: int64
```

In [14]:
```
# drop null values
df.dropna(inplace=True)
```

In [15]:
```
# change data type
df['Amount'] = df['Amount'].astype('int')
```

In [16]:
```
df['Amount'].dtypes
```

Out[16]:
```
dtype('int64')
```

In [17]:
```
df.columns
```

Out[17]:
```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount'],
      dtype='str')
```

In [18]:
```
#rename column
df.rename(columns= {'Marital_Status':'Shaadi'})
```

Out[18]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>User_ID</th> <th>Cust_name</th> <th>Product_ID</th> <th>Gender</th> <th>Age Group</th> <th>Age</th> <th>Shaadi</th> <th>State</th> <th>Zone</th> <th>Occupation</th> <th>Product_Category</th> <th>Orders</th> <th>Amount</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>...</th> <th>11246</th> <th>11247</th> <th>11248</th> <th>11249</th> <th>11250</th> </tbody> <tbody></tbody>

1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare
1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt
1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile
1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction
1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing
...	...	...	...	...	...	...	...	...	..
1000695	Manning	P00296942	M	18-25	19	1	Maharashtra	Western	Chemica
1004089	Reichenbach	P00171342	M	26-35	33	0	Haryana	Northern	Healthcare
1001209	Oshin	P00201342	F	36-45	40	0	Madhya Pradesh	Central	Textile
1004023	Noonan	P00059442	M	36-45	37	0	Karnataka	Southern	Agriculture
1002744	Brumley	P00281742	F	18-25	19	0	Maharashtra	Western	Healthcare

11239 rows × 13 columns

In [19]:
```
# describe() method returns description of the data in the DataFrame (i.e. count
df.describe()
```

Out[19]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>User_ID</th> <th>Age</th> <th>Marital_Status</th> <th>Orders</th> <th>Amount</th> </thead> <tbody> <th>count</th> <th>mean</th> <th>std</th> <th>min</th> <th>25%</th> <th>50%</th> <th>75% </th> <th>max</th> </tbody> <tbody></tbody>

1.123900e+04	11239.000000	11239.000000	11239.000000	11239.000000
1.003004e+06	35.410357	0.420055	2.489634	9453.610553
1.716039e+03	12.753866	0.493589	1.114967	5222.355168
1.000001e+06	12.000000	0.000000	1.000000	188.000000
1.001492e+06	27.000000	0.000000	2.000000	5443.000000
1.003064e+06	33.000000	0.000000	2.000000	8109.000000
1.004426e+06	43.000000	1.000000	3.000000	12675.000000
1.006040e+06	92.000000	1.000000	4.000000	23952.000000

In [20]:
```
# use describe() for specific columns
df[['Age', 'Orders', 'Amount']].describe()
```

Out[20]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Age</th> <th>Orders</th> <th>Amount</th> </thead> <tbody> <th>count</th> <th>mean</th> <th>std</th> <th>min</th> <th>25%</th> <th>50%</th> <th>75%</th> <th>max</th> </tbody> <tbody></tbody>

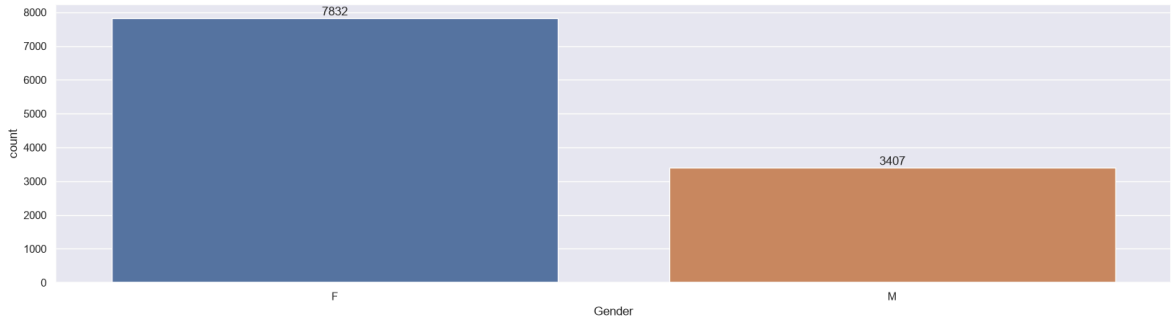
11239.000000	11239.000000	11239.000000
35.410357	2.489634	9453.610553
12.753866	1.114967	5222.355168
12.000000	1.000000	188.000000
27.000000	2.000000	5443.000000
33.000000	2.000000	8109.000000
43.000000	3.000000	12675.000000
92.000000	4.000000	23952.000000

In [21]:
```
### Gender # Exploratory Data Analysis
```

In [52]:
```
# plotting a bar chart for Gender and it's count

ax = sns.countplot(x='Gender', data=df, hue='Gender')

for bars in ax.containers:
    ax.bar_label(bars)
```

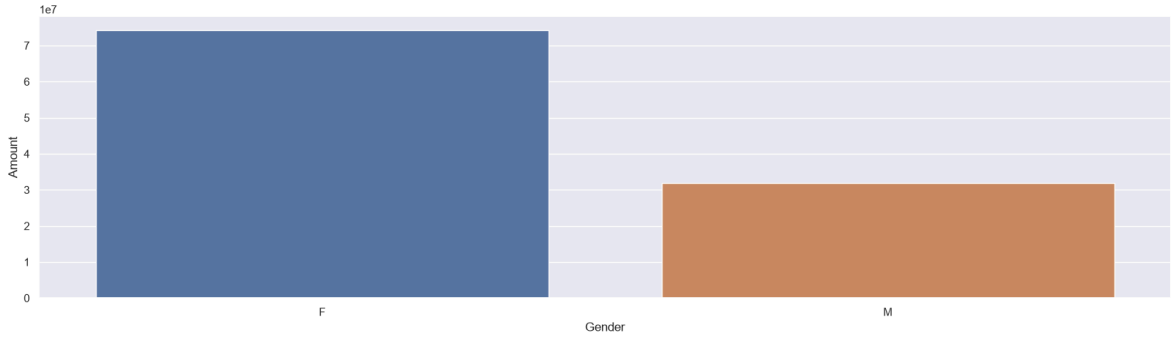


In [49]:
```
# plotting a bar chart for gender vs total amount

sales_gen = df.groupby(['Gender'], as_index=False)['Amount'].sum().sort_values(b

sns.barplot(x = 'Gender',y= 'Amount' ,data = sales_gen)
```

Out[49]: <Axes: xlabel='Gender', ylabel='Amount'>

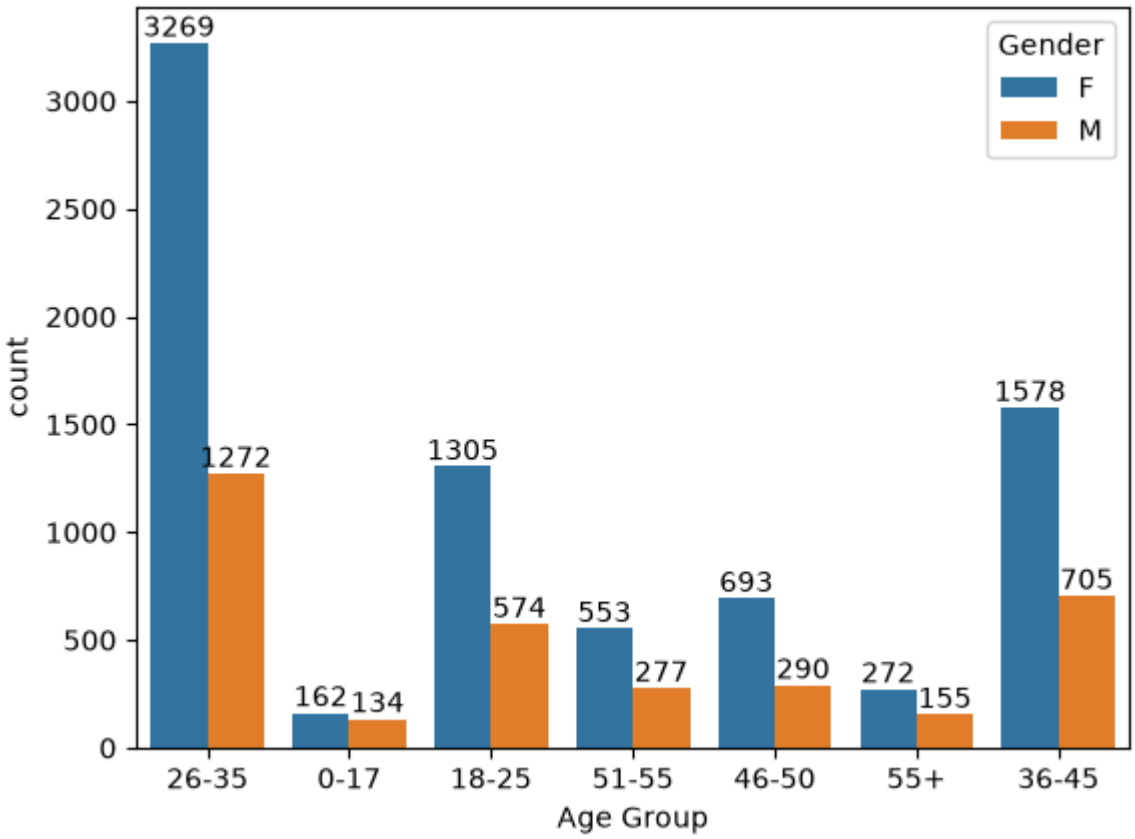


In [ ]:

In [25]:
```
### Age
```

In [26]:
```
ax = sns.countplot(data = df, x = 'Age Group', hue = 'Gender')

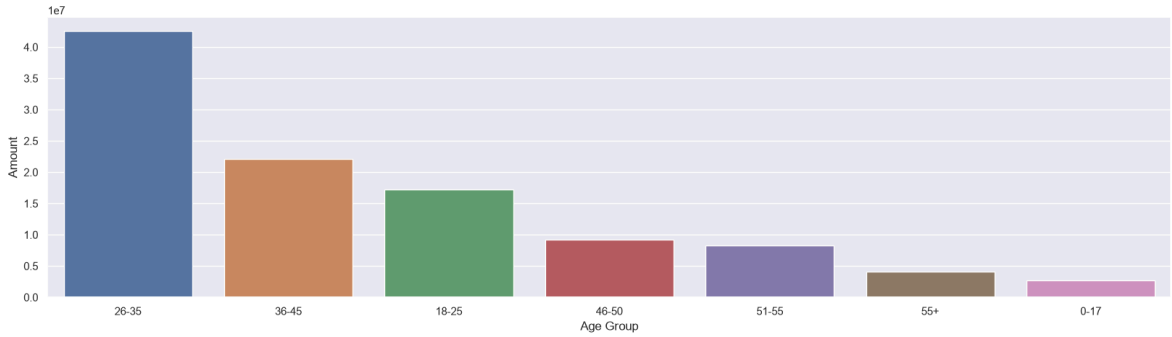
for bars in ax.containers:
    ax.bar_label(bars)
```



In [54]:
```
# Total Amount vs Age Group
sales_age = df.groupby(['Age Group'], as_index=False)['Amount'].sum().sort_value

sns.barplot(x = 'Age Group',y= 'Amount' ,data = sales_age,hue='Age Group', legen
```

Out[54]: <Axes: xlabel='Age Group', ylabel='Amount'>



In [ ]:
```
*From above graphs we can see that most of the buyers are of age group between 2
```

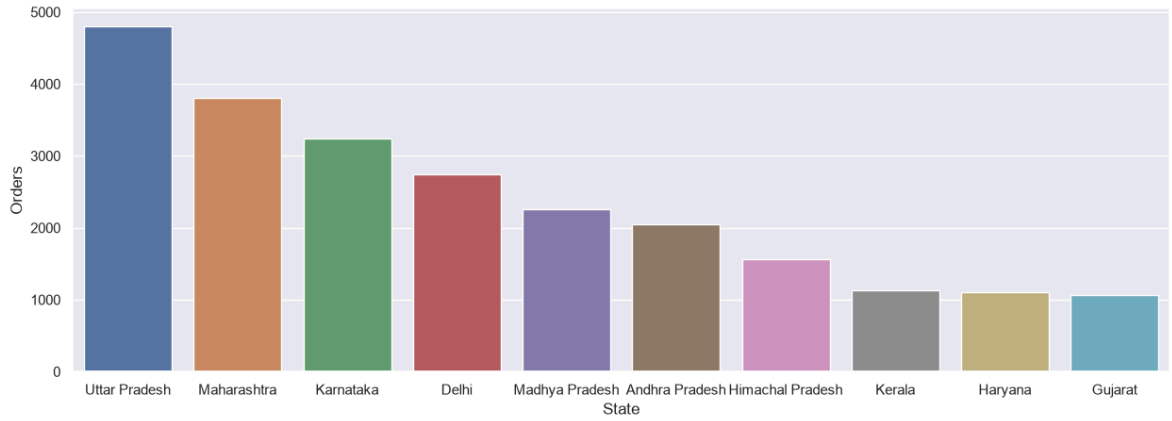
In [ ]:
```
### State
```

In [55]:
```
# total number of orders from top 10 states

sales_state = df.groupby(['State'], as_index=False)['Orders'].sum().sort_values(

sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(data = sales_state, x = 'State',y= 'Orders',hue='State', legend=Fals
```

Out[55]: <Axes: xlabel='State', ylabel='Orders'>

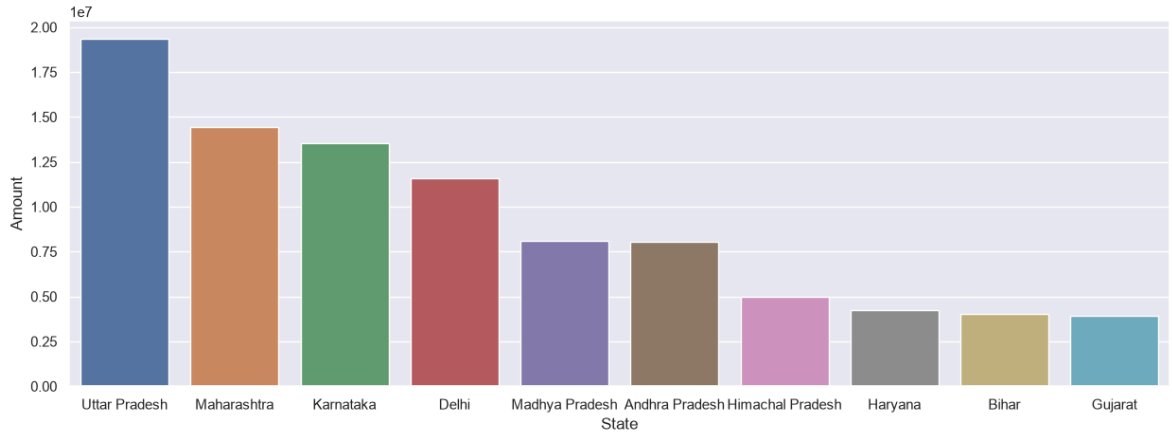


In [56]:
```
# total amount/sales from top 10 states

sales_state = df.groupby(['State'], as_index=False)['Amount'].sum().sort_values(

sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(data = sales_state, x = 'State',y= 'Amount',hue='State', legend=Fals
```

Out[56]: <Axes: xlabel='State', ylabel='Amount'>

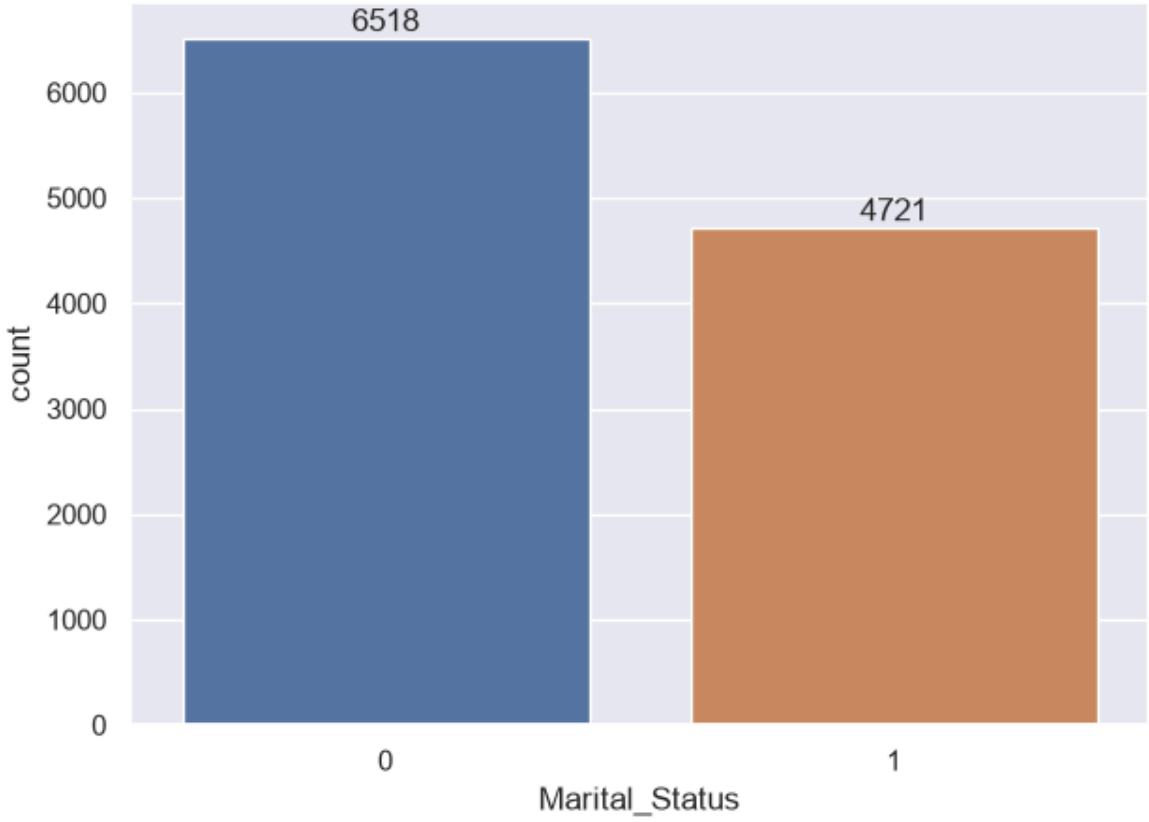


In [ ]:
```
*From above graphs we can see that most of the orders & total sales/amount are f
```

In [32]:
```
    ### Marital Status
```

In [57]:
```
sns.set(rc={'figure.figsize':(7,5)})
ax = sns.countplot(data=df, x='Marital_Status', hue='Marital_Status', legend=Fal

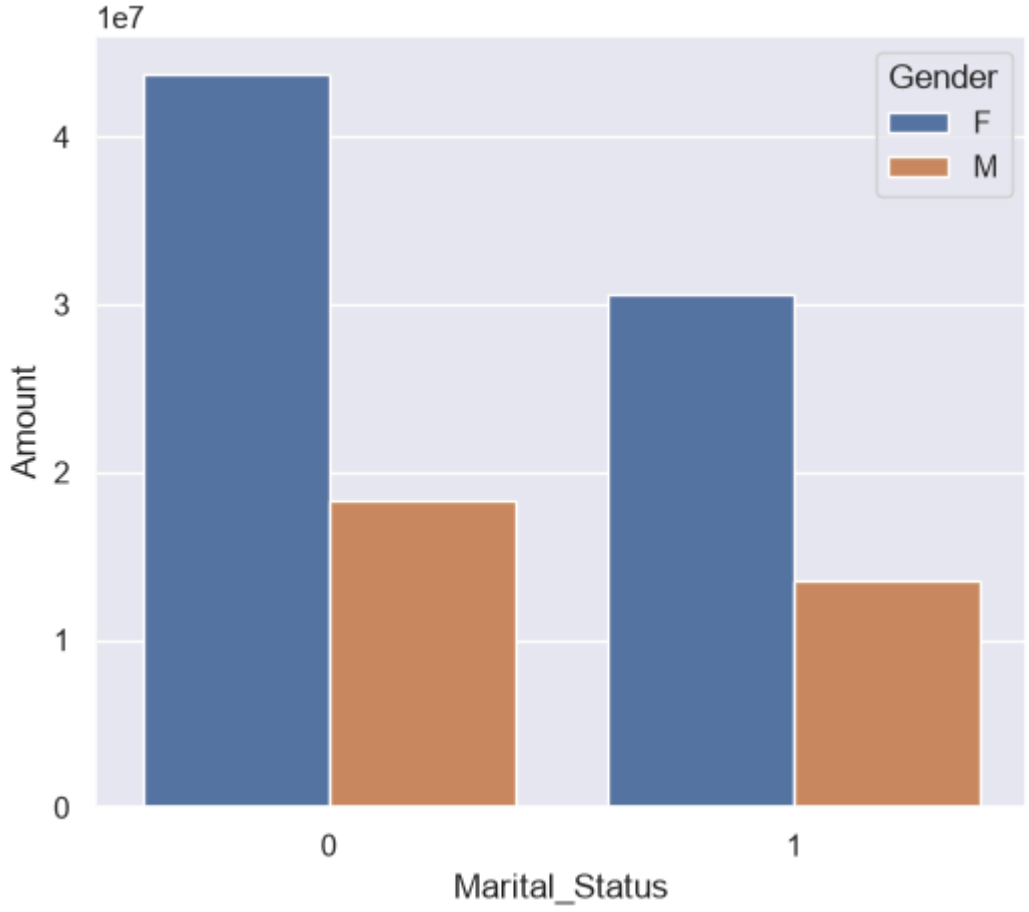
for bars in ax.containers:
    ax.bar_label(bars)
```



In [34]:
```
sales_state = df.groupby(['Marital_Status', 'Gender'], as_index=False)['Amount'].sum()

sns.set(rc={'figure.figsize':(6,5)})
sns.barplot(data = sales_state, x = 'Marital_Status',y= 'Amount', hue='Gender')
```

Out[34]: <Axes: xlabel='Marital_Status', ylabel='Amount'>

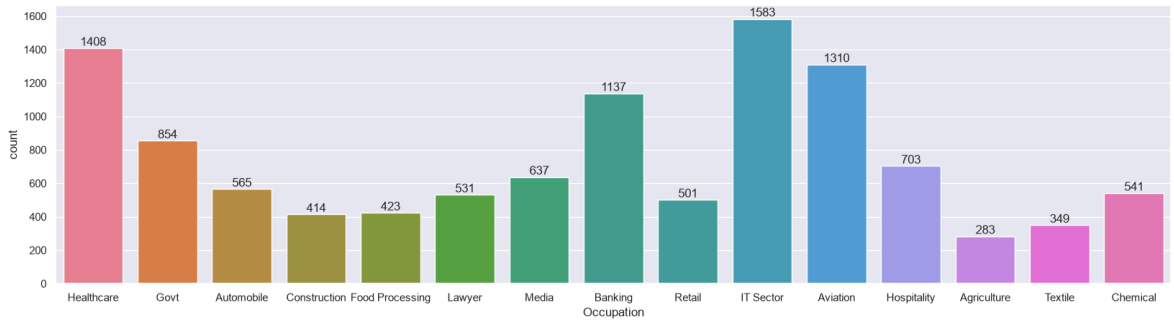


In [ ]:
```
*From above graphs we can see that most of the buyers are married (women) and th
```

In [35]:
```
### Occupation
```

In [58]:
```
sns.set(rc={'figure.figsize':(20,5)})
ax = sns.countplot(data = df, x = 'Occupation',hue='Occupation', legend=False)

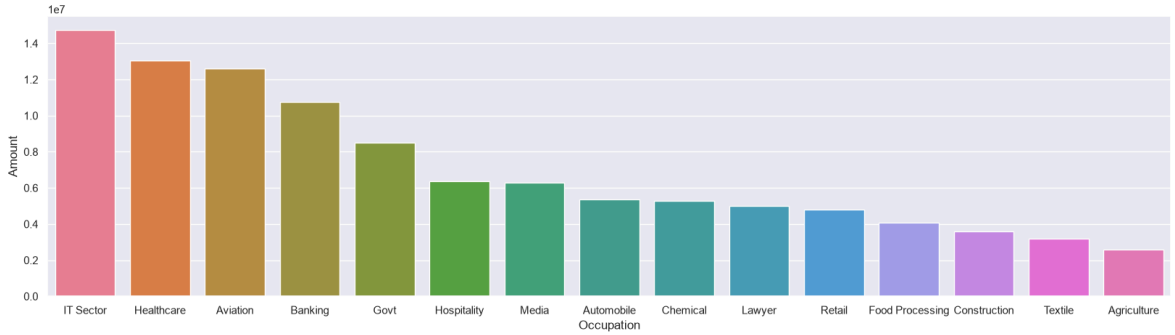
for bars in ax.containers:
    ax.bar_label(bars)
```



In [59]:
```
sales_state = df.groupby(['Occupation'], as_index=False)['Amount'].sum().sort_va

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Occupation',y= 'Amount',hue='Occupation', l
```

Out[59]:
```
<Axes: xlabel='Occupation', ylabel='Amount'>
```

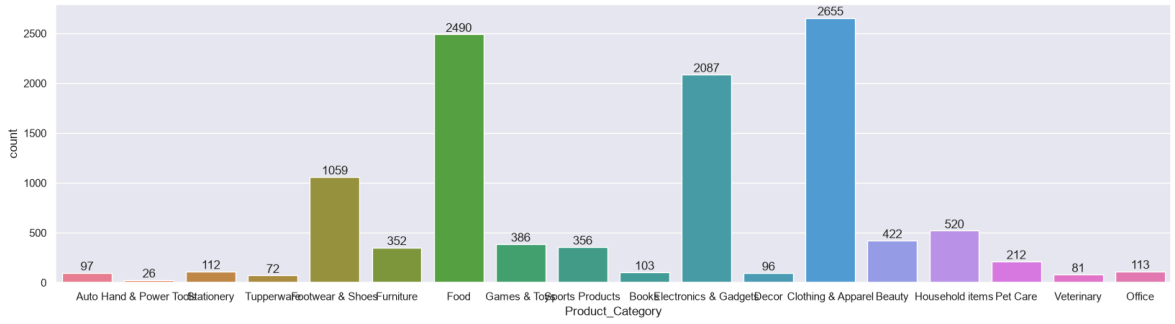


In [ ]:
```
*From above graphs we can see that most of the buyers are working in IT, Healthc
```

In [38]:
```
### Product Category
```

In [60]:
```
sns.set(rc={'figure.figsize':(20,5)})
ax = sns.countplot(data = df, x = 'Product_Category',hue= 'Product_Category',leg

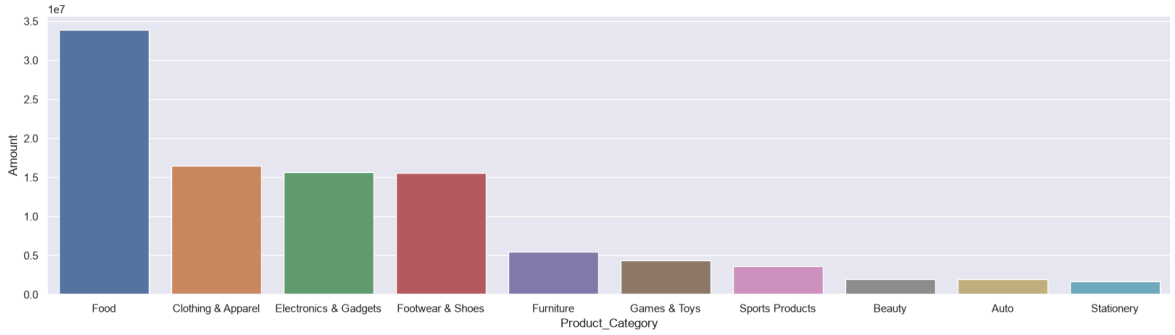
for bars in ax.containers:
    ax.bar_label(bars)
```



In [61]:
```
sales_state = df.groupby(['Product_Category'], as_index=False)['Amount'].sum().s

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_Category',y= 'Amount',hue= 'Product
```

Out[61]:
```
<Axes: xlabel='Product_Category', ylabel='Amount'>
```

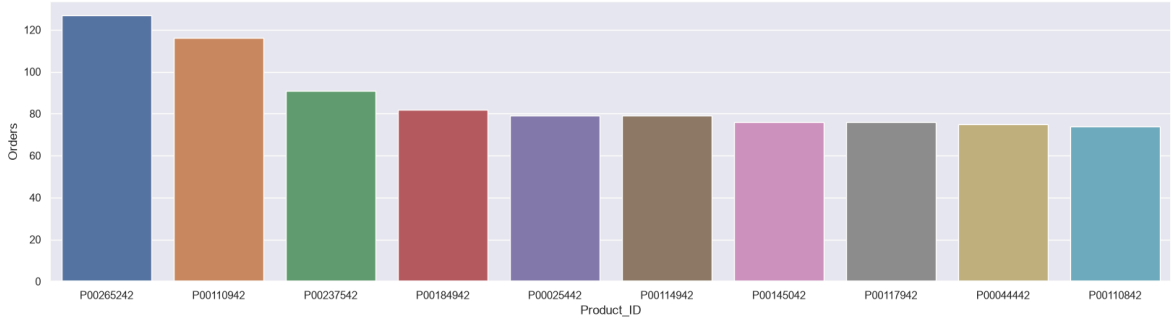


In [ ]:
```
*From above graphs we can see that most of the sold products are from Food, Clot
```

In [62]:
```
sales_state = df.groupby(['Product_ID'], as_index=False)['Orders'].sum().sort_va

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_ID',y= 'Orders',hue= 'Product_ID',l
```

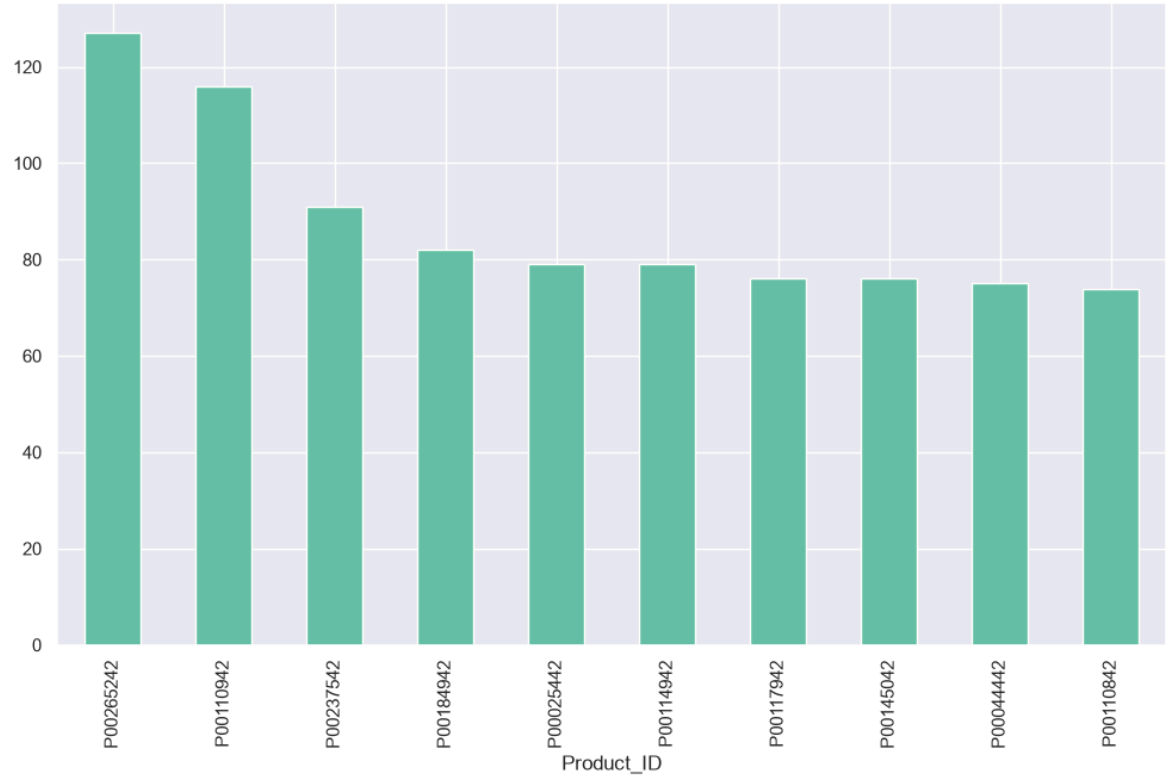
Out[62]:
```
<Axes: xlabel='Product_ID', ylabel='Orders'>
```



In [65]:
```
# top 10 most sold products (same thing as above)

fig1, ax1 = plt.subplots(figsize=(12,7))
df.groupby('Product_ID')['Orders'].sum().nlargest(10).sort_values(ascending=Fals
```

Out[65]: <Axes: xlabel='Product_ID'>



In [44]:

In [ ]:
```
*Married women age group 26-35 yrs from UP,  Maharastra and Karnataka working in
```